

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Attempt the following questions.

[12]

A) a) Use the following statements

Apply

p: Mohan is rich

q: Mohan is happy

write the following statement in symbolic form

- i) Mohan is rich but unhappy.
- ii) Mohan is poor but happy.
- iii) Mohan is neither rich nor happy.

b) Construct the truth table for the $p \vee \neg(p \wedge q)$ statement form.

B) a) Write the following statements using quantifier variables and predicate symbols Understand

- i) All birds can fly.
- ii) Some men are genius.
- iii) Each integer is either even or odd.

b) In survey of 120 people, it was found that 65 read News read magazine, 45 read Times, 42 read Fortune; 20 read both Network and Times, 25 read both Network and Fortune, 15 read both Time and Fortune; and 8 read all the three magazine. Find the number of people who read exactly one magazine.

Q.2 Solve Any Two of the following.

[12]

A) Let $A = \{1, 2\}$ and $B = \{a, b, c\}$ Find

Understand

- i) $A \times B$
- ii) $B \times A$
- iii) $A \times A$.

B) Solve $a_r = a_{r-1} - 6a_{r-2} = -30$ given $a_0 = 20$, and $a_1 = 5$.

Apply

C) a) Let R be the relation on the set of real numbers such that xRy if and only if x and y are real numbers that differ by less than 1, that is $|x - y| < 1$. Show that R is not an equivalence relation.

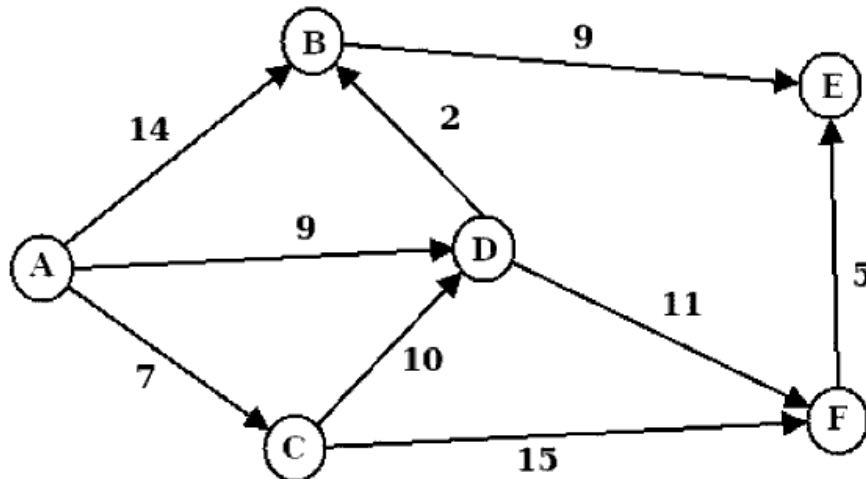
b) Define composite function. Let f and g be the functions from the set of

integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$.
What is the composition of f and g ? What is the composition of g and f ?

Q. 3 Solve Any Two of the following.

[12]

- A) Define the terms: Simple Path, Null Graph, Complete Graph, Planner Graph. Find the Hamiltonian path and Hamiltonian circuit in the complete graph $K_{4,3}$. Understand
- B) Compute the shortest distance between source A to destination E using Dijkstra's algorithm for the following graph. Understand



- C) i. Write a Handshaking Lemma for a graph having Vertices $V = (G, E)$ and Edges E . Understand
- ii. How many edges are there in a graph with 10 vertices each of degree six?
- iii. Define Chromatic Number with suitable example.

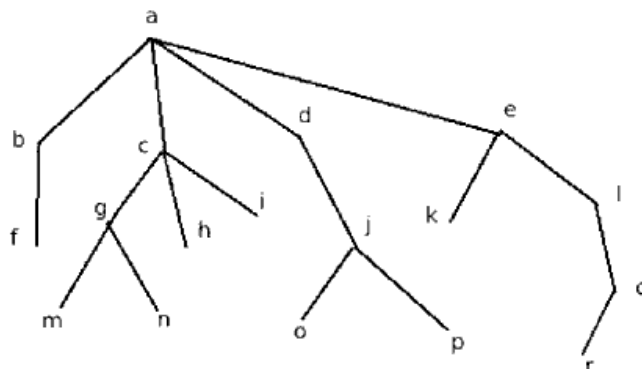
Q.4 Attempt the following questions.

[12]

- A) Consider the following rooted tree and give the answer for following:

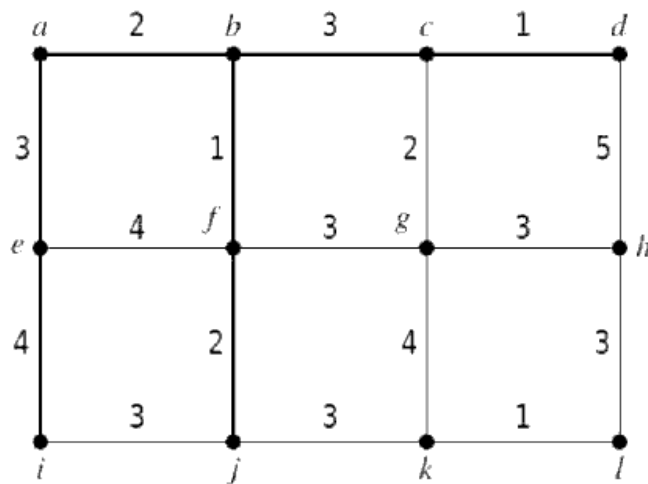
Understand

- (i) Find the ancestor of f .
- (ii) Find the Descendant of j .
- (iii) How many terminal vertices are there?
- (iv) Draw the subtree rooted at e .
- (v) Find the siblings of g .
- (vi) Write the internal vertices of the tree.



- B) Use Prim's algorithm and Kruskal's to find a minimum spanning tree in the graph shown in Figure.

Apply



Q. 5 Attempt the following questions.

[12]

- A) Consider the binary operation defined on the set $A = \{a, b, c, d\}$ by following table. Understand
Find:

*	a	b	c	d
a	a	c	b	d
b	d	a	b	c
c	c	d	a	a
d	d	b	a	c

- (i) $C * d$ and $d * c$
(ii) $b * d$ and $d * b$
(iii) $a * (b * c)$ and $(a * b) * c$
- B) Consider the group $G = \{1, 2, 3, 4, 5, 6\}$ under multiplication modulo 7. Understand
i) Find multiplication table of G .
ii) Find 2^{-1} , 3^{-1} .
iii) Find the orders and subgroups generated by 2 and 3.

*** End ***